The Islamia University of Bahawalpur

Department of INFORMATION TECHNOLOGY

Faculty of Computing

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

for

< Online Doctor Appointment Application >

Version 1.0

By
ZARNAB FATIMA
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Supervisor

SIR MUSTAFA HAMEED

Revision History

Name	Date	Reason for changes	Version

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	Dr.	Supervisor, SENG- 4301	<date>

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken

Supervised by

SIR MUSTAFA HAMEED

Signature _____

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1. Introduction

The intended audience for this document includes all stakeholders involved in the Online Doctor Appointment Application (ODAA) project, such as administrative staff, doctors, patients, and developers. It is recommended that they refer to this document for the official and up-to-date requirements. Any requirements mentioned elsewhere, either in written or verbal form, should not be considered valid unless they appear in this document or its revisions.

1.1 **Overview**

This section provides a general description of the Online Doctor Appointment Application (ODAA) and its purpose.

1.2 **User Characteristics**

Describes the different types of users who will interact with the ODAA and their roles, such as administrative staff, doctors, and patients.

1.3 **General Constraints**

Specifies any constraints or limitations that need to be considered during the development of the ODAA.

1.4 **Assumptions**

States any assumptions made during the requirements gathering process and their impact on the ODAA.

User and Application Requirements

This section outlines the specific requirements of the ODAA, including both functional and nonfunctional requirements.

Design Documentation and Implementation Decisions

Provides an overview of the design documentation and implementation decisions made for the ODAA.

Test Cases

Presents a set of test cases to verify and validate the functionality of the ODAA.

Progress and Challenges

Summarizes the progress made during the development of the ODAA and discusses any challenges faced during the process.

Conclusion

Concludes the Software Requirements Specification (SRS) document and emphasizes its importance as the primary source of requirements for the project.

1.1 Purpose

The purpose of this document is to describe the requirements for the Online Doctor Appointment Application (ODAA). The intended audience includes all stakeholders in the potential system, including administrative staff, doctors, patients, and developers. Developers should consult this document and its revisions as the sole source of requirements for the project. They should not consider any requirements statements, whether written or verbal, as valid until they appear in this document or its subsequent revisions.

1.2 Scope

The scope of the Online Doctor Appointment Application (ODAA) project is to design and develop a fully functional online platform for booking doctor appointments. The application will enable customers to book appointments with doctors from anywhere in the world through the application. Doctors can register on the application, view their appointments, and set their available time slots for appointments, making it more convenient for them. The administrator will have access to the application and will be able to modify its information and access the database. Payment for appointments will be made after the appointment has taken place.

1.3 Definitions, Acronyms, and Abbreviations.

Table : 1. **Design and implementation constraints**

Term	Definition
Appointment	An arrangements to meet someone at a particular time and place.
Android Application	An application run on the internet with Mobile device
Fire-base	This is Data Base and record store
Constraints	Illumination of application.
Business Domain	A set of classes that represents object in the business model being implemented.
Self-contained	Having all that is needed by it self.
Administrative staff	Person responsible for management of operations.
Graphic user interface	A visual way of interacting with a android using items.
Software Requirements Specification	A document that completely describes all of the functions of a proposed application and constraints under which must operate.
Stakeholder	Any person with an interest in the project who is not developer
User	Patient, Doctor and Admin

1.4 References

Google Scholar Research Papers

And Something Search from Google engine

1.5 Overview

It's important to note that the scope of the ODAA project is focused on the development of the online doctor appointment application. Other aspects such as medical diagnosis, treatment, or prescriptions are beyond the scope of this project. The application will comply with relevant health care regulations and standards and will be developed using suitable technologies compatible with the target platform and infrastructure.

The key features and functionalities within the scope of the ODAA project include:

Customer Features:

Registration and account creation for customers.

Search functionality to find doctors based on various criteria.

Detailed doctor profiles with information on qualifications and patient reviews.

Appointment booking with selected doctors based on availability.

Confirmation notifications for appointments.

Doctor Features:

Registration and profile creation for doctors.

Management of availability and setting preferred appointment slots.

View and management of appointments, including rescheduling or cancellations.

Access to patient information and medical history with appropriate consent and privacy measures.

Notifications of new appointments and any changes made by patients.

Administrator Features:

Access to the application's back-end for system management.

Addition and updating of doctor profiles, including specialties and availability.

Management of customer accounts and administrative tasks. Database

access for maintenance and reporting purposes.

2. The Overall Description

This Software Requirements Specification (SRS) is the requirements work product that formally specifies Online Doctor Appointment Application (ODAA). It includes the results of both business analysis and systems analysis efforts. Various techniques were used to elicit the requirements and we have identified your needs, analyzed and refined them. The objective of this document therefore is to formally describe the system's high level requirements including functional requirements, non-functional requirements and constraints.

The detail structure of this document is organized as follows:

First we will provide an overview of the business domain that the proposed Online Doctor Appointment Application (ODAA) will support. These include a general description of the product, user characteristics, general constraints, and any assumptions for this Application. Second we will mention the User and Application

Requirements. After that, we will take a look at the design documentation and implementation decisions of the Application and some test cases. At the end, my report will conclude the progress I made and discuss the problems faced during the work.

2.1 Product Perspective

This Doctor Appointment Application is a self-contained application that allows patients to book appointment and doctors to manage appointments. Various stakeholders are involved in this application.

2.2 Product Functions

a. Provide an application that enables patients to book an appointment with any available doctor.

b. Doctors will be able to view their appointments and manage them properly.

c. Coordinate various calendars and finding available time slots for appointment.

d. Reserve equipment and rooms for appointment.

e. Alert patients in case there is an earlier available time slot.

2.3 User Characteristics

The system will be used in the hospital. The doctors and patients will be the main users. Given the condition that any person who use just android cell phone Some users may have to be trained on using the application. The system is also designed to be userfriendly. It uses a Graphical User Interface (GUI).

Patient: These are the people who want to make the appointment.

Doctor: These are the specialist whom appointments are being booked for.

Database Administrator: They are responsible for maintaining and overseeing the fire-base of the application.

2.4 General Constraints

>> The system must be delivered by the proposed deadline.

>>The system must be user-friendly.

2.5 Assumptions and Dependencies

>> It is assumed that Android device will be available before the application is installed and tested.

>> It is assumed that the Hospital will have enough trained staff to take care of the application.

3. Specific Requirements

Design and implementation constraints

It is an android application in which we want to design the feature to solve the problem of awareness of this type of technology in the urban areas and improve the medical field. In which we implement the low cost budget so people can easily enjoy the app. One thing is most important at the people to implement the medical treatment within the time.

Table:2. **Requirements of trace-ability Matrix**

Tools	**Version**	**Rationales**
VS Code	1.74.0 (latest)	IDE
Adobe Illustrator	2020	Icon/Logo Work
XAMPP	8.1.12 (latest)	Local server environment for running web software
Diagram.io	2023	Gant Chart
Technology	**Version**	**Rationales**
HTML	-	Used for creating the structure of the web
CSS/ Bootstrap	5.3.0 (latest)	styling
PHP	8.1.0 (latest)	Programming language
Laravel	10.2	PHP framework for developing web apps

jQuery	3.6.0	JavaScript framework for DOM
API	_	For cloud service integration
SQL	_	Used for database operations
JavaScript	ECMAScript	Used for event-driven programming

3.1 Functional Requirements

3.2.1 < ***Functional Requirement or Feature*** >

REQ1. The application should enable patients and doctors to log in.

REQ2. The application should enable patients and doctors to register.

REQ3. The application should enable patients and doctors to log out.

REQ4. The application should allow Patients to book appointment. .
REQ5. The application should allow Patients to search for available doctors.

REQ6. The application should allow Patients to cancel their appointment.

REQ7. The system should allow Patients and Doctors to view and modify their profile.

REQ8. The system should allow Doctors to set their available time.

REQ9. The system should enable administrator to log in.

REQ10. The system should allow administrator to manage users .
REQ11. The system should allow administrator to delete past appointments.

REQ12. The system should allow administrator to manage & access Fire-base.

3.2 Data Flow Chart Diagram

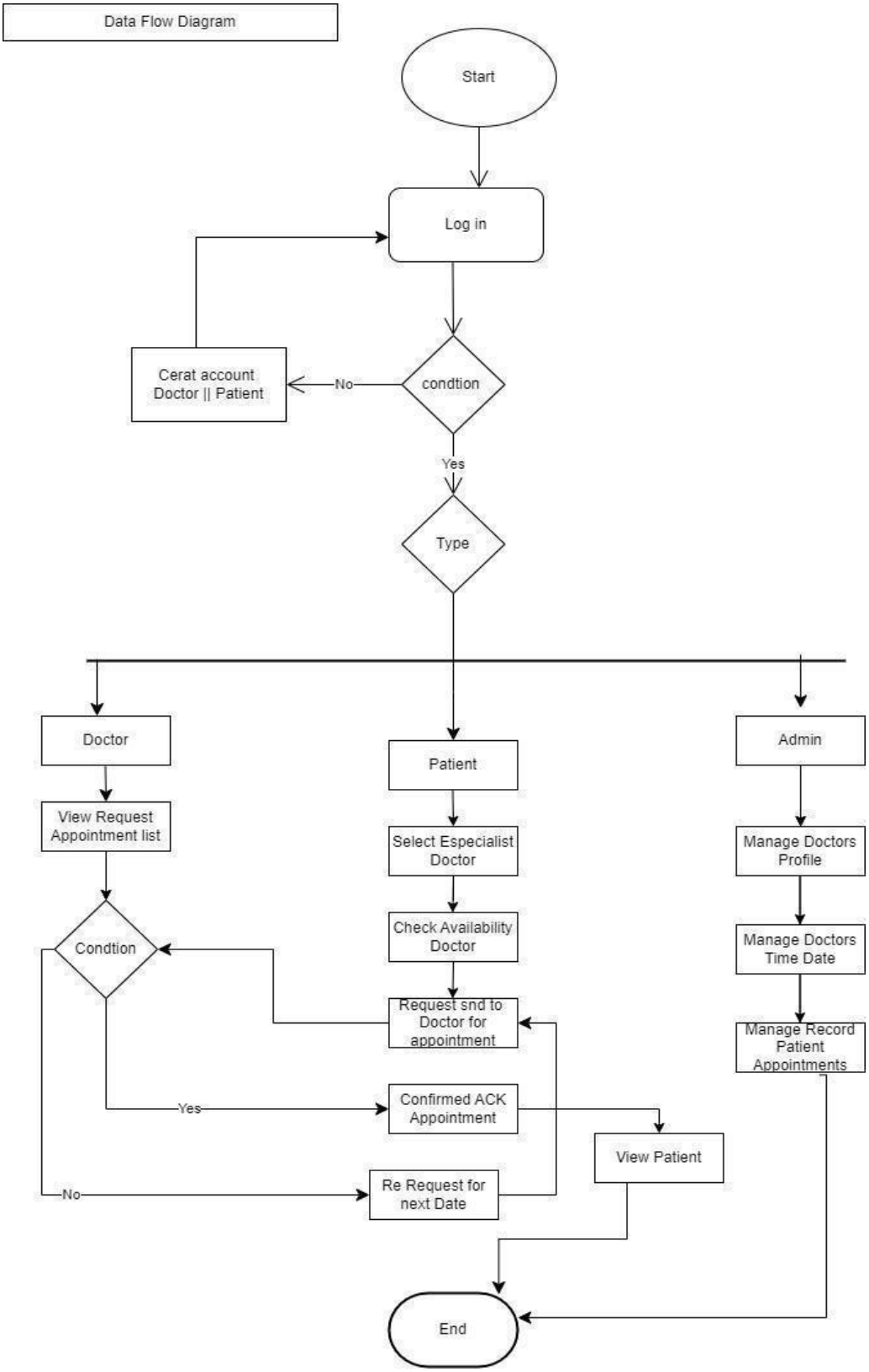

Figure 1. Data Flow Chart diagram for Doctor Appointment Application

3.3 Use Cases

3.3.1 #1 **use case diagram for Doctor Appointment Application**

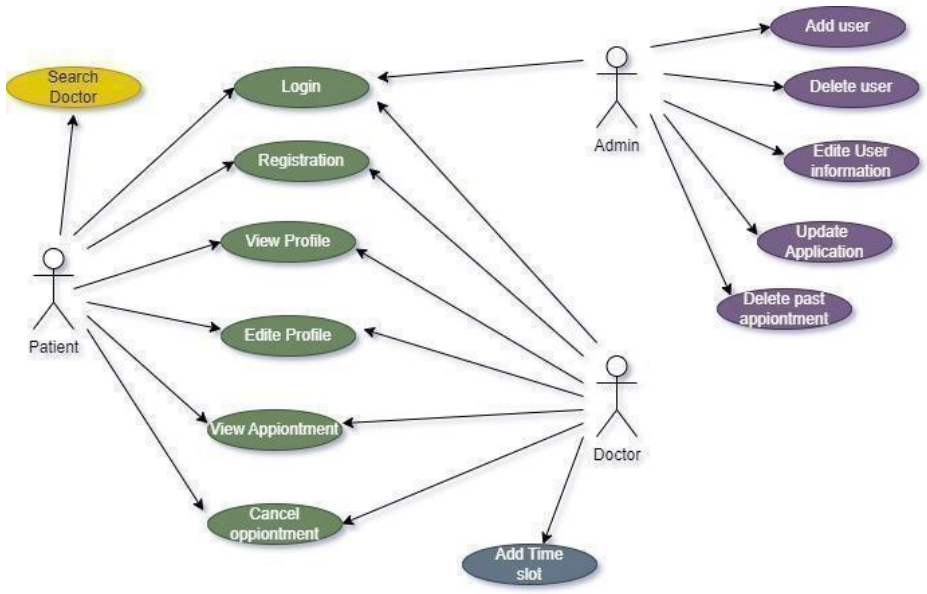

Figure: 2 Use case diagram for Doctor Appointment Application

3.3.2 #2 ***use case diagram for Doctor registration & Login***

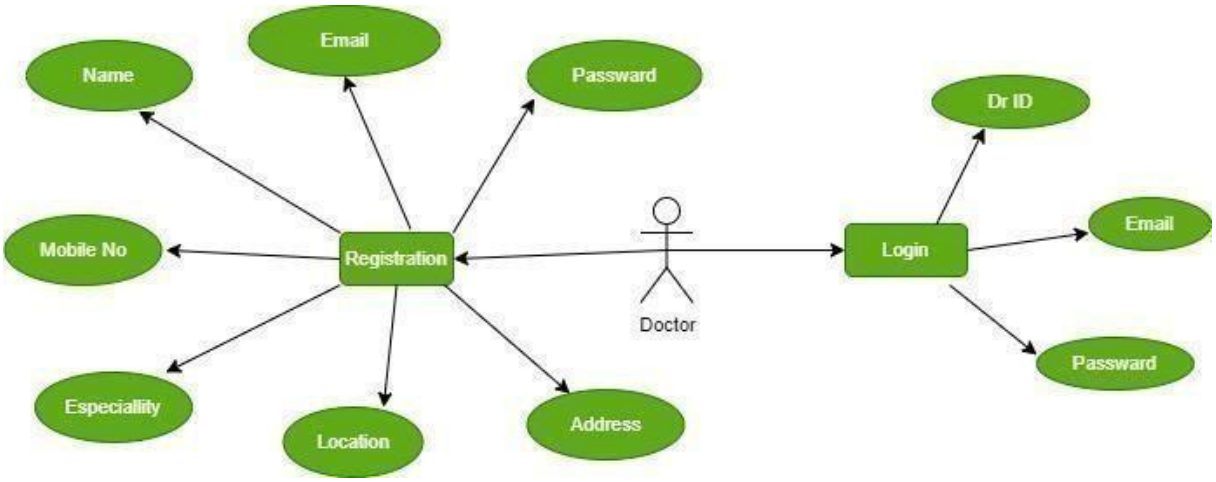

Figure 3. Use case diagram for Doctor registration & Login

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3.3.3 #3 **use case diagram for Doctor registration & Login**

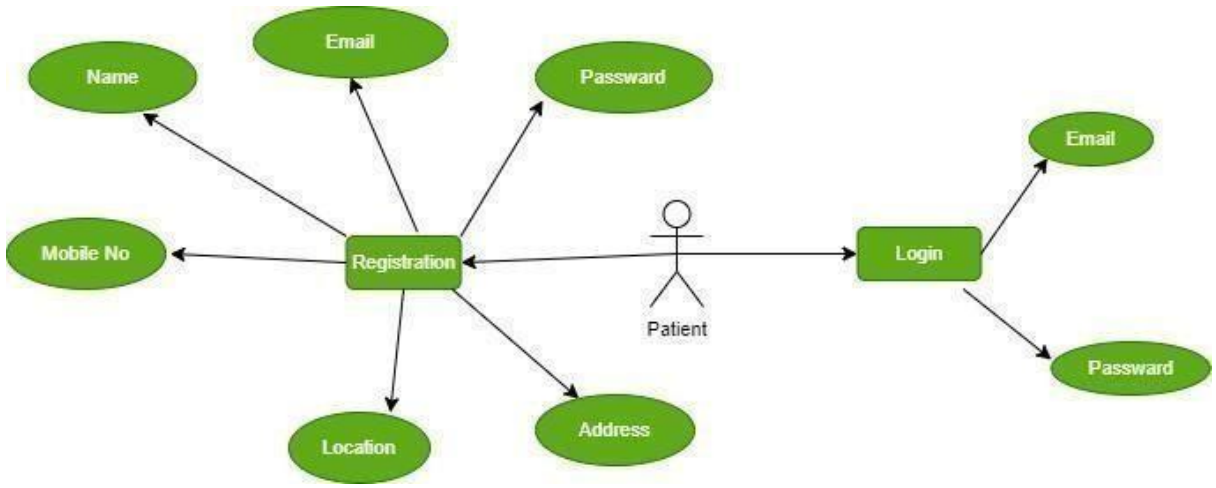

Figure 4 use case diagram for Patient registration & Login

Detailed Use Case Specification

Table : 3 Use Case for Login specification

Use Case UC-1:	Login
Related Requirements:	REQ1, REQ3, REQ4,REQ6, REQ7, REQ8, REQ9, REQ10, REQ11, REQ12
Initiating Actor:	Patient, Doctor, Database administrator
Actor's Goal:	To log into account
Participating Actors:	
Preconditions:	Application displays the menu for doctor or patient to enter username and password.

Flow of Events for Main Success Scenario:

Table : 4 Main Scenario for login

	1.	Doctor/patient opens app and clicks Log in.
	2.	Application shows user a menu to enter username and password.
	3.	User enters the information and click on submit button.
	4.	Application displays “login successful”

Table : 5 Use Case for Register specification

Use Case UC-2:	Register
Related Requirements:	REQ2, REQ4, REQ7, REQ8, REQ9
Initiating Actor:	Patient, Doctor
Actor’s Goal:	To Create an account that can book appointment for patient and for doctor to able confirm and manage appointment.
Participating Actors:	
Preconditions:	The system displays the menu for doctor or patient to enter personal information.

Flow of Events for Main Success Scenario:

Table : 6 Main Scenario for Register

	1.	Doctor/student open App and clicks register to create an account
	2.	System shows user a page to get some basic information about user such as name, surname, user ID, date of birth, username, password and etc.
	3.	User enters the information and click OK to account to be made
	4.	Application displays “login successful”

3.4 Classes Diagram

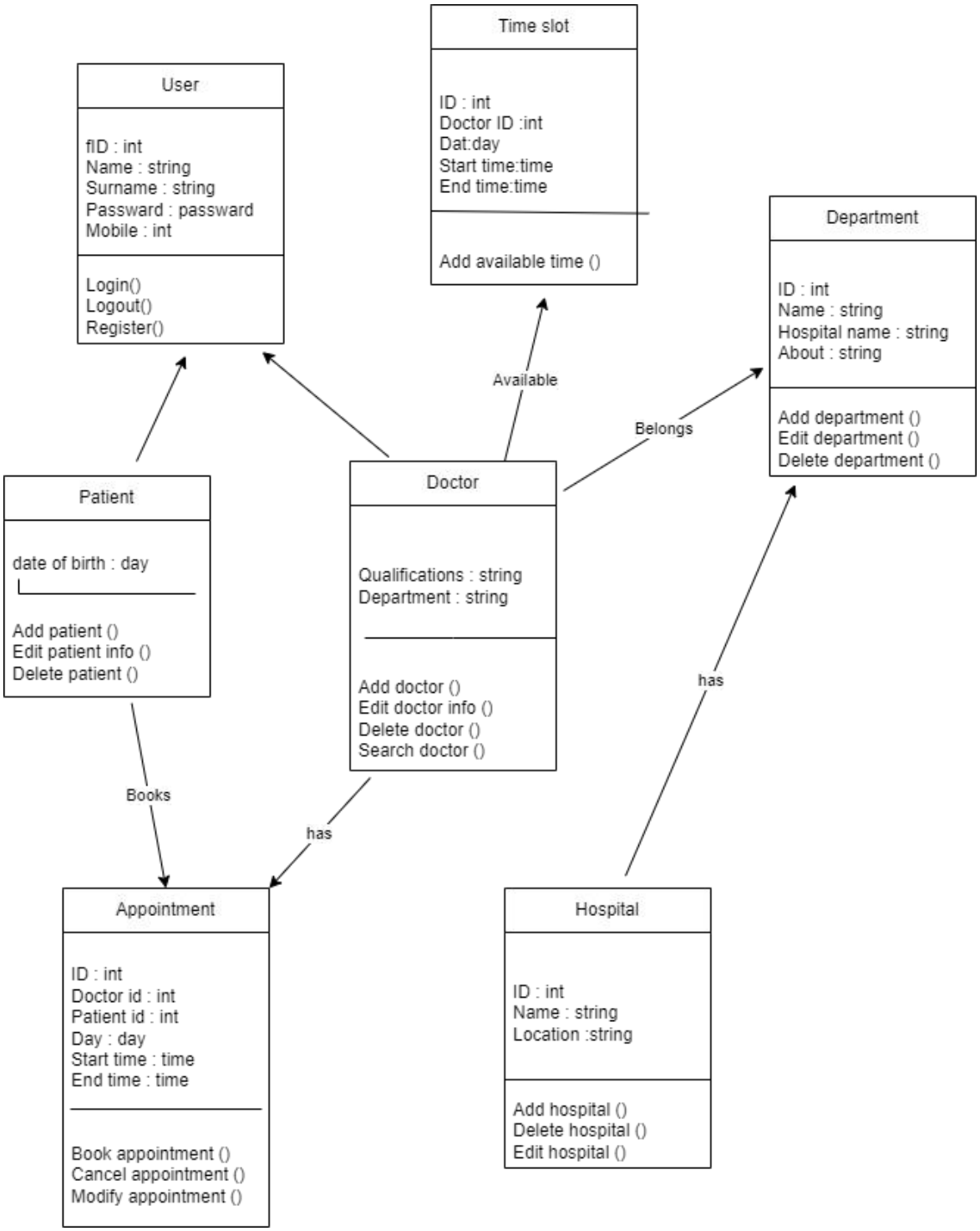

Figure 5. Class diagram for Doctor Appointment Application

3.5 Non-Functional Requirements

i. Reliability

The application should be available when requested for service by users.

The application should have a very low failure rate.

ii. Performance

The application must have a good response time.

The application should be able to achieve a lot in a specified amount of time.

The application must run error free while operating with a huge set of data.

iii. Security

All external communications between the application and clients must be encrypted.

The access permissions for application data may only be changed by the application data administrator.

All application data must be backed up every 24 hours and the backup copies stored in a secure location which is not in the same building as the application.

iv. Usability

The application should include well-structured user manuals.

The application should have Informative error messages. Efficient help facilities.

The application should have a well-formed graphical user interfaces.

The application should be user friendly.

v. Safety

The application should maintain a good back up.

vi. Support-ability

The application should be able to be transferred from one environment to another.

The application should be easy to maintain.

The application should be able to deal with additional international conventions such as languages, or number formats, styles.

The application should be able to be used on multiple platforms.

3.6 System Requirements

1. functional requirements

User Login:

The application should enable patients and doctors to log in.

Users should enter their username and password.

The entered information should be validated.

Access should be granted or denied based on the provided credentials.

User Registration:

The application should enable patients and doctors to register.

For patients, collect user information such as names, date of birth, address, telephone, email, and password.

For doctors, collect information such as names, email, password, date of birth, gender, department, address, and telephone.

Check the validity of the information provided:

Password field should not be empty.

Password and confirm password fields should match.

Email should not have been used before.

If the information is valid, save and add the user to the database (Firebase).

User Logout:

The application should enable patients and doctors to log out.When a user clicks on the log out button, they should be logged out of the application.

Appointment Booking:

The application should allow patients to book appointments.

The application should check if the patient is logged in.

Patients should select the department and hospital of interest.

The application should display the available time slots of the selected doctor for booking.
The application should generate a unique booking reference for each appointment.
A confirmation email should be sent to the patient after making the appointment.

The application should require the patient's email for sending a response.

A comment field should be provided for the patient to describe any specific issues or concerns.

Doctor Search:

The application should allow patients to search for available doctors of a particular field.

Patients should be able to search by entering the first name or last name of the doctor.

The application should display all doctors that match the patient's search criteria.

The available time slots of the doctors should be displayed.

Appointment Cancellation:

The application should allow patients to cancel their bookings.

Patients should be able to modify reservations without re-entering all their information.

Patients only need to provide their booking reference for making changes.

The necessary updates should be made in the system after modifications.

Profile Management:

The application should allow patients and doctors to view and modify their profiles.

Users should be able to enter new information that replaces the old information in the database (Fire-base).

Doctor's Availability:

The application should allow doctors to set their available time.

Doctors should be able to enter the time slots when they will be available.

This information should be saved in the database (Fire-base).

Administrator Login:
The application should enable the administrator to log in.

The administrator should enter their username and password.
The provided information should be validated.

Access should be granted or denied based on the provided credentials.

User Management by Administrator:

The application should allow the administrator to manage users.

The administrator should be able to access the database and add new customers.

The administrator should be able to delete users from the database based on predefined rules.

The administrator should be able to change patient or doctor information, such as last name, email, password, department, etc.

After the administrator writes a reply, it should be sent to the user.

Delete Past Appointments:

The application should allow the administrator to delete past appointments from the system.

After the date of an appointment has passed, the administrator should delete the appointment from the database.Database Management by Administrator: **2. Non-functional Requirements**

i.Reliability

The application should be available when requested for service by users: The system should work 24/7, it should always be up and running so that whenever the user wants to use it, it's available.

The application should have a very low failure rate: The failure rate should be kept as minimal as possible, preferably less than 0.01.

ii. Performance

> The application must have a good response time.

> The load time for the user interface should take less than two seconds.

> The log in information should be verified within five seconds.
> The application should be able to achieve a lot in a specified amount of time.

> The application should be able to withstand a heavy workload.

> It should be able to respond to multiple numbers of people at the same time.

> The application must run error free while operating with a huge set of data.

> The application should be precise and accurate when dealing with data.

> The application error rate should be minimal.

iii. Security

All external communications between the application data server and clients must be encrypted:

To ensure that the application is secure access will be protected by a user log in screen and requires a user name and password. The access permissions for application data may only be changed by the application data administrator: The application administrator should be the only one with the authority to enable access to the application data.

All application data must be backed up every 24 hours and the backup copies stored in a secure location which is not in the same building as the application: This is done to avoid loss of information in case of application crash.

iv. Usability

The application should include well-structured user manuals.

The application should have a well- structured easy to understand manual to guide its users.

The application should have Informative error messages.

It should explain what the user did wrong.

It should show where exactly the error can be found.

It should explain how to recover from the error.

The error message should be simple to understand.

The application should have a well-formed graphical user interfaces.

The application should be user friendly:

The application must be easy to learn for both novices and users with experience from similar application.
The application must be efficient for the frequent user.

The application must be easy to remember for the casual user.

The user must understand what the application does.

The user must feel satisfied with the application.

v. Safety

The application should maintain a good backup: Maintaining backups ensures that the application fair base is secured, which means that in case of an emergency or accident the application can be easily restored.

vi. Support ability

The application should be able to be transferred from one environment to another.

The application should still work perfectly when it is transferred from one operating environment to another.

It should run on Android devices.

The application should be easy to maintain.

In other for the application to be easy to maintain it should be done with an object oriented language which is easy to maintain.

The application should be able to be used on flutter.

4. Analysis Models

Analysis models are used to describe the requirements and behavior of a system. Here are some analysis models that are relevant to a Online Doctor Appointment Application:

1. **Use case model**: A use case model describes the interactions between the application and its users, and identifies the different use cases or scenarios in which the application will be used. In a Online Doctor Appointment Application.
2. **Data model**: A data model describes the data that is used by the Application, including the entities (such as Doctor, Patient, and Appointment) and the relationships between them. In a Online Doctor Appointment Application.

4.1 Sequence Diagrams

#1 **Sequence diagram for Doctor & Patient Login**

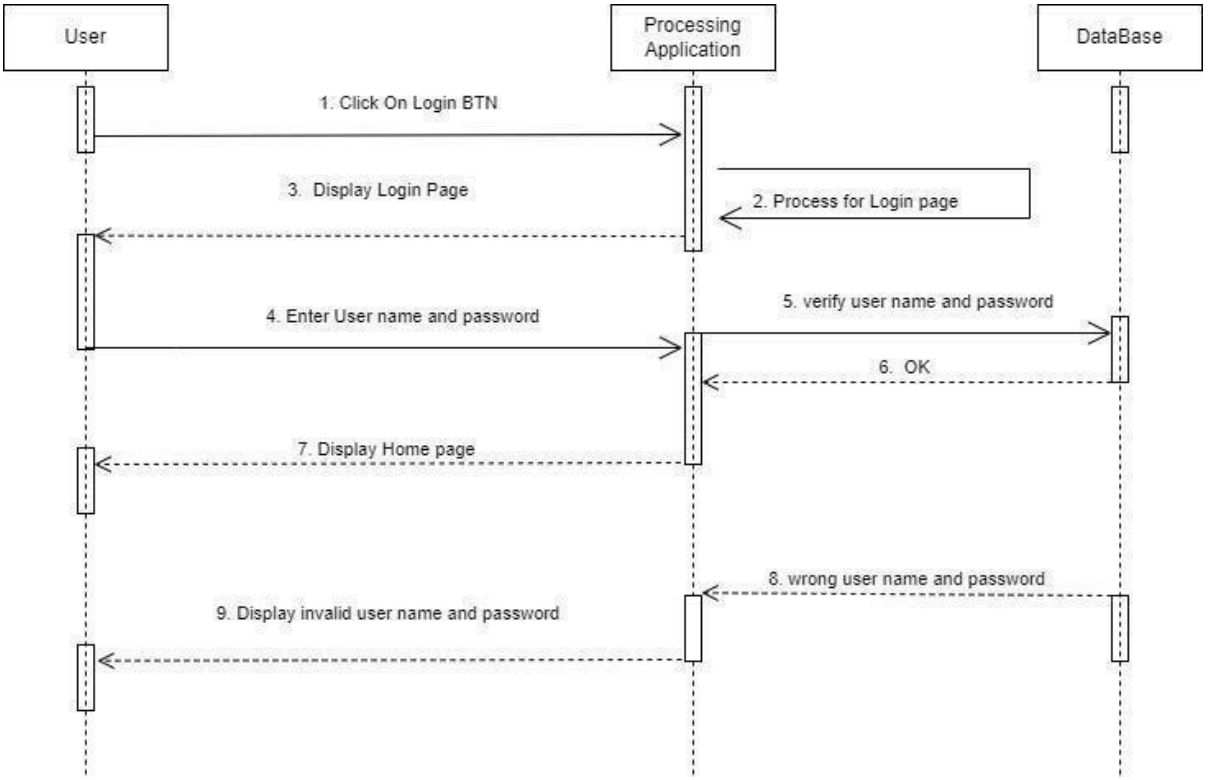

Figure 6. Sequence diagram for Doctor & Patient Login

#2 **Sequence diagram for Doctor & Patient Registration**

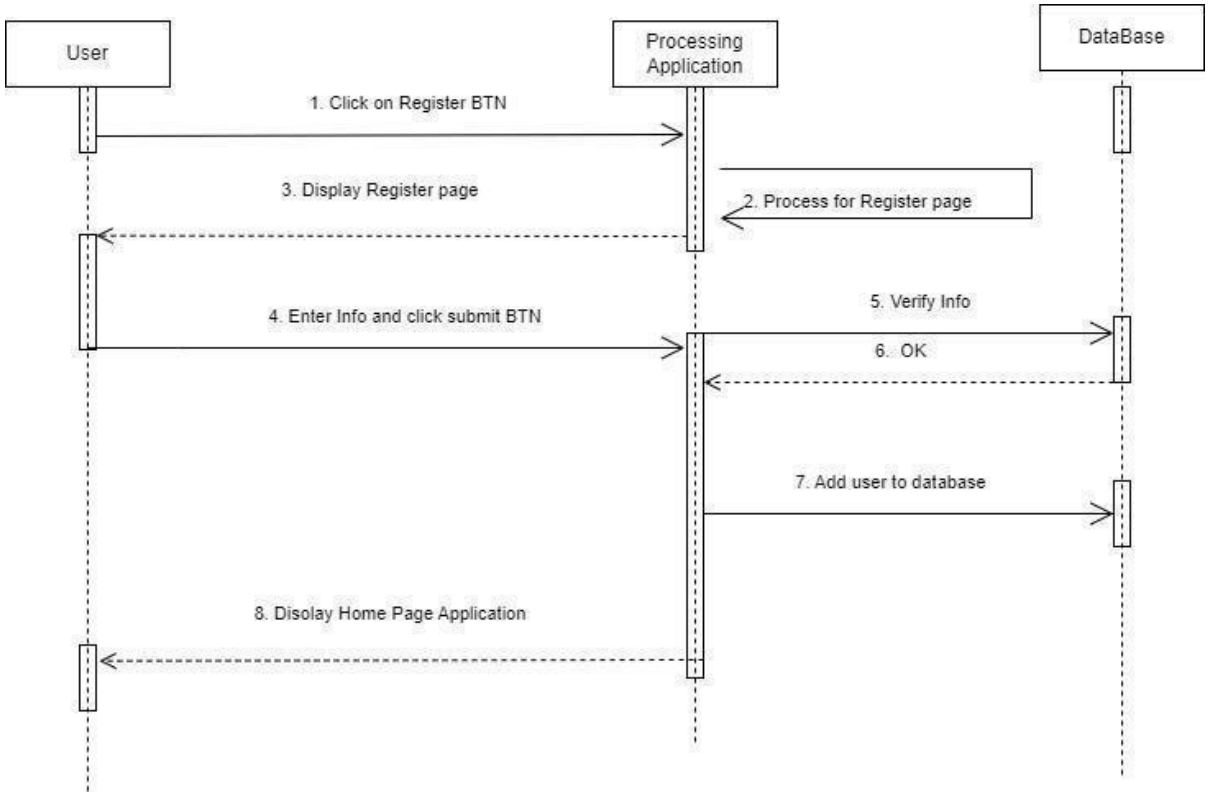

Figure 7. Sequence diagram for Doctor & Patient Registration

#3 **Sequence diagram for Patient Book Appointment**

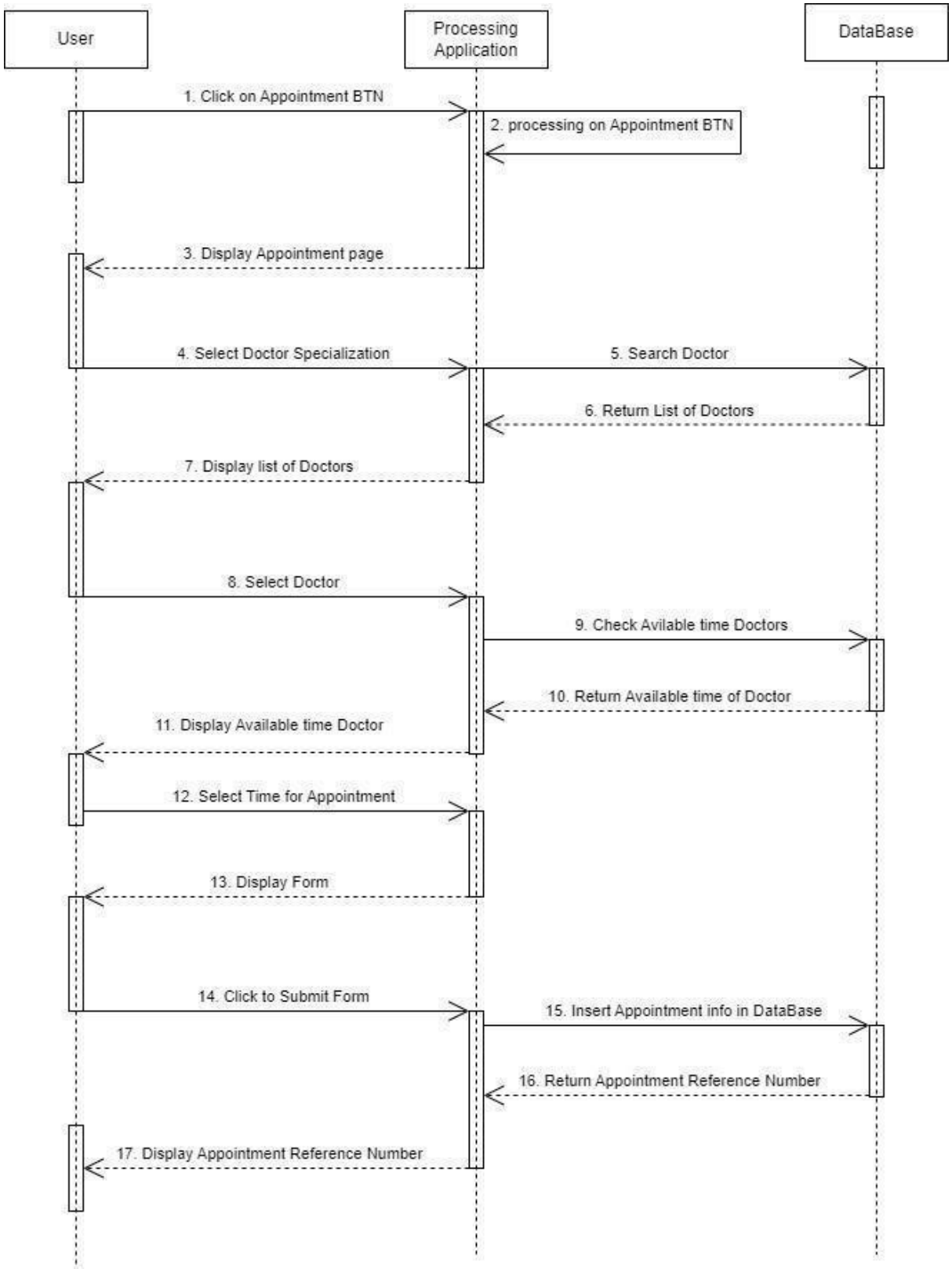

Figure 8. Sequence diagram for Patient Book Appointment

4.2 Data Flow Diagrams (DFD)

Data Flow diagram for Online Doctor Appointment Application

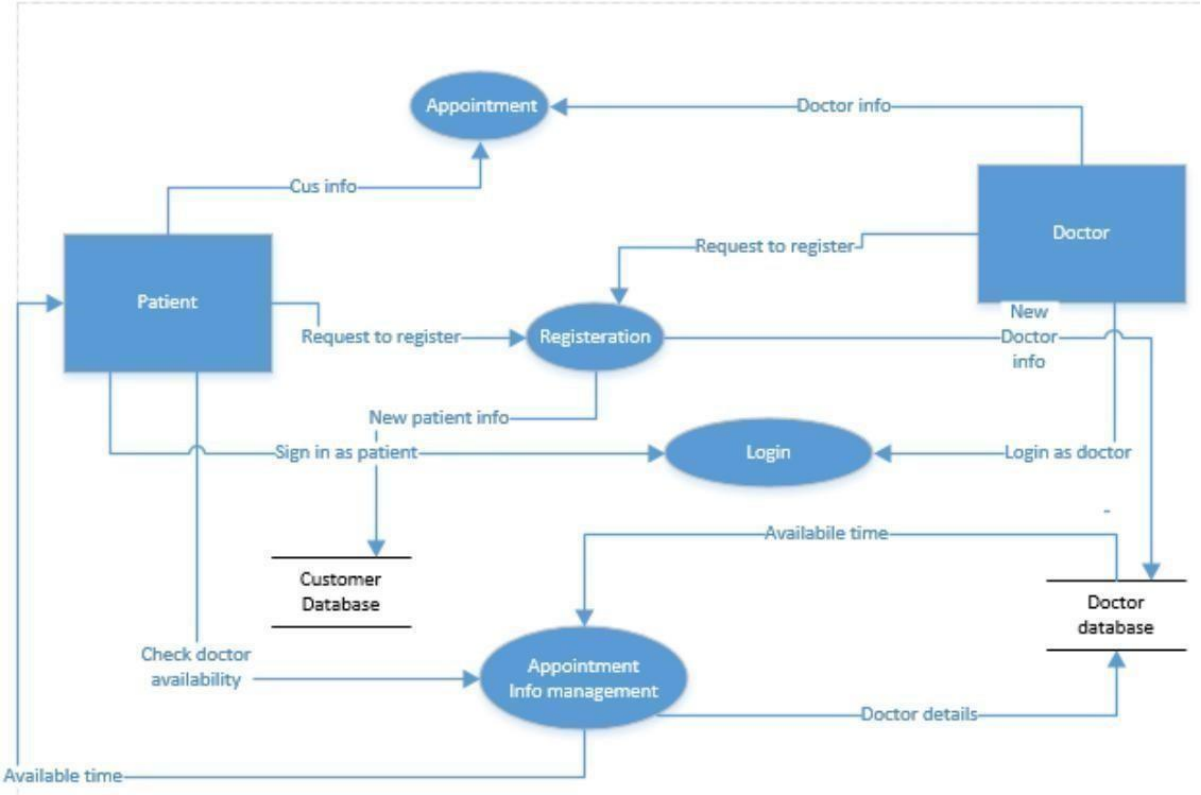

Figure 9. Data Flow diagram for Online Doctor Appointment Application

4.3 State-Transition Diagrams (STD)

#1 **STD For Doctor & Patient Login**

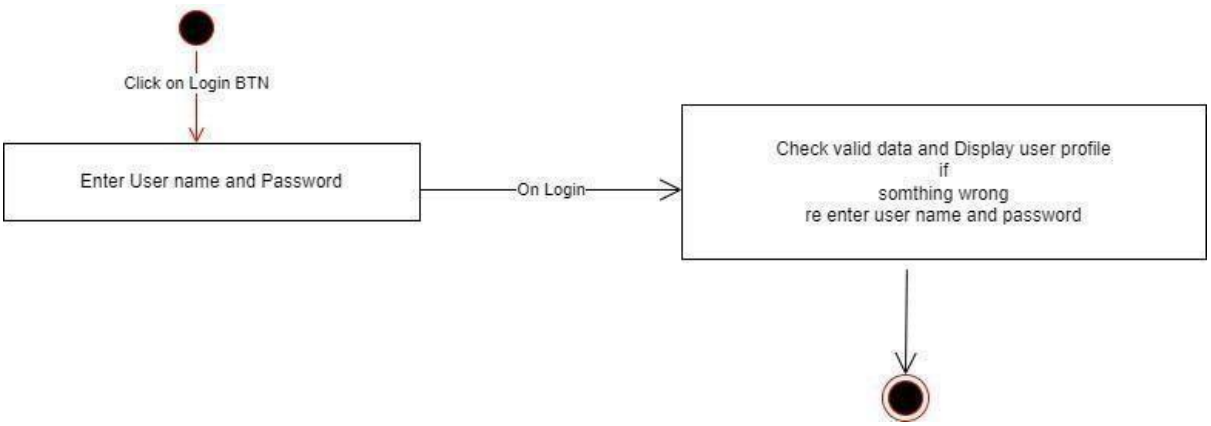

Figure 10. STD for Doctor & Patient Login

#2 **STD For Doctor & Patient Registration**

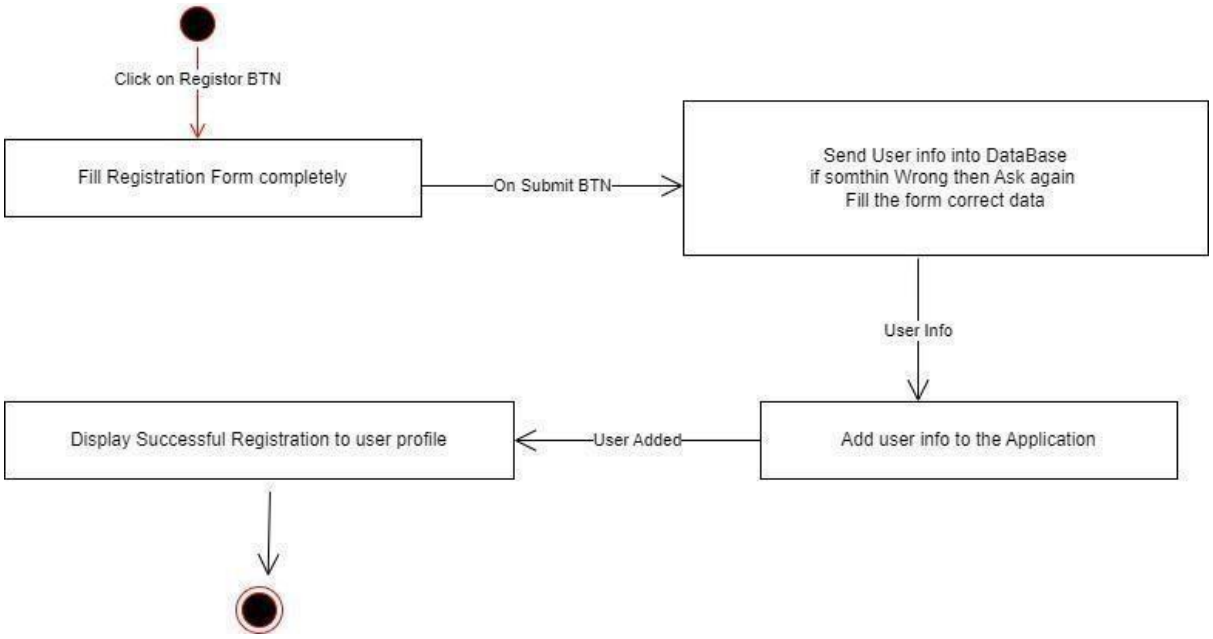

Figure 11. STD for Doctor & Patient Registration

#3 **STD For Patient Book Appointment**

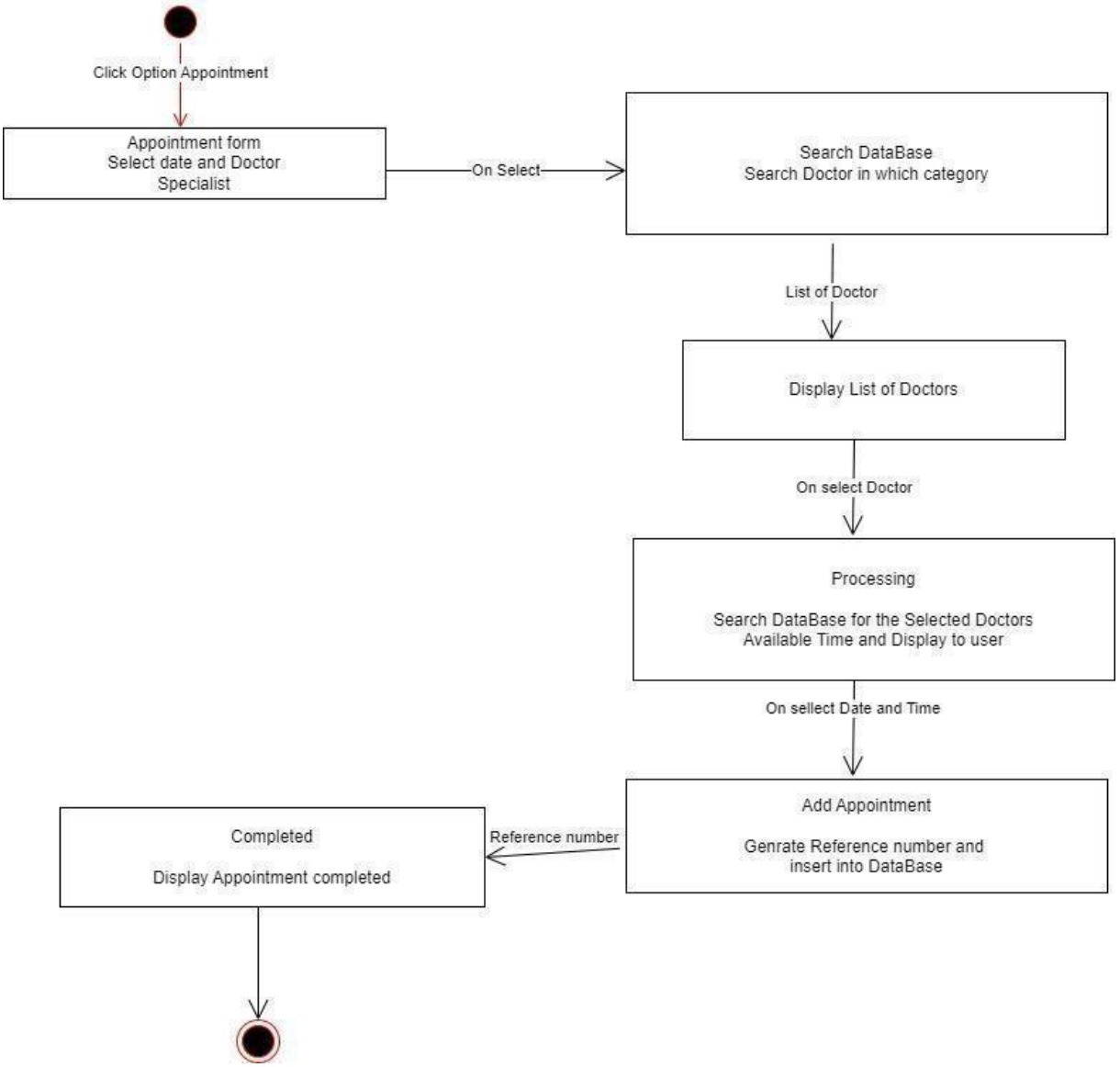

Figure 12. STD for Doctor & Patient Registration

5. Data Dictionary

Table : 7 **Admin &User:**

Name	Description	Data Type	Length	Null
ID	Admin ID	Integer	5	NO
Name	Admin Name	String	80	NO
Address	Admin Address	String	100	NO
MobileNo	Mobile Number	String	12	NO
Username	Username for Login	String	40	NO
Password	Password for Login	Integer	8	NO
ID	Admin ID	Integer	5	NO

Name	Description	Data Type	Length	Null
User ID	User ID	Integer	5	NO
Name	User Name	String	80	NO
Address	User Address	String	100	NO
Mobile No	Mobile Number	String	12	NO

Table : 8 **Login:**

Username	Username for Login	String	40	NO
Password	Password for Login	Integer	8	NO